Optimized Bus Route Finder for Educational Institutes using Haversine formula

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***Abstract*— The following paper delves into the innovative integration of technology into the bus transportation system provided by education institutes. The paper explores the user-centric features offered by the system, aiming to enhance the overall commuting experience for passengers. The webapp serves as a comprehensive platform, providing users with detailed information about various bus routes, driver details for each route, and schedules for different time slots, including morning, afternoon shuttle, and evening buses. The standout feature of the system is its advanced geospatial functionality, allowing users to effortlessly locate the nearest bus stop based on the latitude and longitude coordinates they provide.**

Keywords—Haversine formula, Google Maps API

I. INTRODUCTION

The webapp categorizes bus services into morning, afternoon shuttle, and evening schedules, providing a comprehensive overview of daily transportation options. The app's core functionalities include providing detailed information about various bus routes, their respective timings, and visual representations of stops on Google Maps. This facilitates ease of navigation for students.

The provision of bus driver details, including names and contact numbers, fosters a sense of familiarity and accountability within the student community. Additionally, information about the bus in-charge, along with their contact details, further contributes to a transparent and communicative transportation system.

The webapp also has the ability to find the nearest stops within the 2 Kilometre radius of the user specified geo co-ordinates.

II. RELATED WORKS

There are many situations where an optimal path or route needs to be found between multiple locations. Many models have been developed by researchers to predict the possible routes and paths from multiple locations. Each and every model use a variety of optimization techniques and routes. The applications for which these optimal route needs to be determined also vary and based on the applications, the models have been built.

The research done by Rezania Agramanisti Azdy and Febriyanti Darnis focuses on proposing a solution to waste management challenges by using the Haversine formula to determine distances between drivers and temporary shelters. Through GPS technology, the coordinates of the locations are obtained and the formula calculates distances between them [1]. The main objective was to optimize the order of visits to temporary shelters, prioritizing those closest to drivers. This approach is expected to result in cost savings and reduced travel time to the waste collection park. In particular, the Haversine formula, known for its accuracy in calculating distances between geographic points, became a key tool in this waste management strategy.

Another study by Vinayak Hegde et al, the Haversine formula was utilized to calculate distance between students' residential address and educational institutes [2]. The paper demonstrated how educational institutes can enhance their marketing strategies by using data mining. By analyzing the distance from students' residential addresses to educational institutions, the proposed system was able to provide more accurate insights into prospective admissions. The Haversine formula calculated orthodromic distance, the shortest distance between two points on the Earth's surface, based on latitude and longitude. The k-means algorithm was employed for clustering locations, and Google Maps API was utilized for visualization. The system predicted admission rates, allowing institutions to target specific areas for improved advertising, ultimately increasing admission rates and customer engagement.

In 2018, Yuda Dian Harja and Riyanarto Sarno conducted research on enhancing medical service accessibility by developing a location-based system that incorporates critical variables like travel distance and time. The Haversine algorithm calculated distances between a user's location and nearby medical facilities within a specified radius. The system further leveraged the Google Maps API to determine

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travel distances and times, providing a more comprehensive evaluation of available medical options. The TOPSIS algorithm was then applied to weigh and rank these options based on multiple criteria, including radius, distance, and time [3]. The experimental findings highlighted the significance of considering various factors rather than just the radius when making decisions about medical service selection. The research outcomes revealed a high degree of agreement of 95 % between the proposed method and human assessments, underscoring the effectiveness of the developed system in aiding users during emergency medical situations.

The research conducted by Anisya, Ganda Yoga Swara, primarily focusses on system designed to address the challenges of earthquake and tsunami evacuation in the city of Padang, which is prone to such disasters due to its geographical location [4]. The purpose of the research was to help the community identify the nearest tsunami shelter by applying a heuristic method to an Android application based on artificial intelligence. The proposed system utilized the best-first-search method to search for the nearest shelter, considering factors such as path length, population density, and travel time. The haversine formula was used to calculate distances between locations. The research focused on overcoming difficulties in selecting evacuation routes, specifically in the red zone (coastal territory), and is limited to data obtained within vulnerable areas or 2 km from the coastline.

This paper uses the haversine formula to find the best possible routes and a bus route number to a particular destination. The model gives more than one possible option with an optimized result.

III. Proposed System

The proposed model finds the possible buses for a particular route. There may be any number of buses or routes that can be availed possible to a particular destination. The system does not list all the possible routes, but is lists only the buses that cover within the 2 Kilometre radius of the user specified geo co-ordinates. And the buses are searched and listed for a particular period of time.

The Morning, Afternoon and Evening components is designed to facilitate user interaction with the bus routes. The working of the developed web interface is explained as follows:

A. Route Selection and Search

Users can select a specific bus route from a dropdown menu. Upon selecting a route, the user can click the "Search" button to retrieve information about the stops and timings associated with that route.

B. Data Retrieval

The component sends requests to the server using Axios to fetch relevant data about morning/afternoon/evening bus routes based on the selected route number.

C. Display of Bus Stops and Timings

Upon successful data retrieval, the component displays a table showing the sequential stop number, stop name, and corresponding timings for each stop on the selected morning bus route.

D. Bus Driver Information

The component fetches and displays information about the bus driver responsible for the selected route, including the driver's name, contact number, in-charge name, and in-charge contact number.

E. Google Maps Integration

The component utilizes the Google Maps API to display a map with markers indicating the locations of each bus stop on the selected route.

Clicking on a map marker shows an information window with details about the specific stop, including its name, timing, and a link to view the location on Google Maps.

The map dynamically adjusts its centre and zoom level based on the latitude and longitude of the first stop on the selected route, ensuring an optimal view of the route.

In the web app a feature called nearest stop finder has been implemented. This uses Haversine formula to calculate the nearest stops from a table which contains all the geo-coordinates of all stops of all routes, based on the user specified geo-coordinates.

The Haversine formula is expressed as follows:

$a = \sin^2((lat2 - lat1)/2) + \cos(lat1) * \cos(lat2) * \sin^2((long2 - long1)/2)$

$c = 2 * \text{Math.atan2}(\text{Math.sqrt}(a) / \text{Math.sqrt}(1-a))$

$d = R * c$

lat1 and long1 are the latitude and longitude of the first point in radians,

lat2 and long2 are the latitude and longitude of the second point in radians,

R is the radius of the Earth (mean radius = 6,371 kilometre's),

a is the square of half the chord length between the points on the sphere,

c is the angular distance in radians,

d is the straight-line distance between the two points on the surface of the sphere.

The stops within a 2 Kilometre radius are found and are displayed to the user in a map.

TABLE I. SAMPLE USER LOCATION

Latitude	**Longitude**
13.090476	80.215826

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TABLE II. DISTANCE CALCULATED USING HAVERSINE FORMULA

Route Number	Stop Name	Latitude	Longitude	Distance in KM
2	Shanti Colony	13.084926	80.213364	0.67227
3	Thirumangalam	13.087033	80.198703	1.89363
AF4	Mambakkam	12.827745	80.165147	29.72604
24	Kotturpuram	13.023194	80.242871	8.03453
22A	Mappedu	12.891023	80.140697	23.62484

TABLE II showcases the distance between the user's given location and all the Stops in the database where Haversine formula is used to calculate the distance.

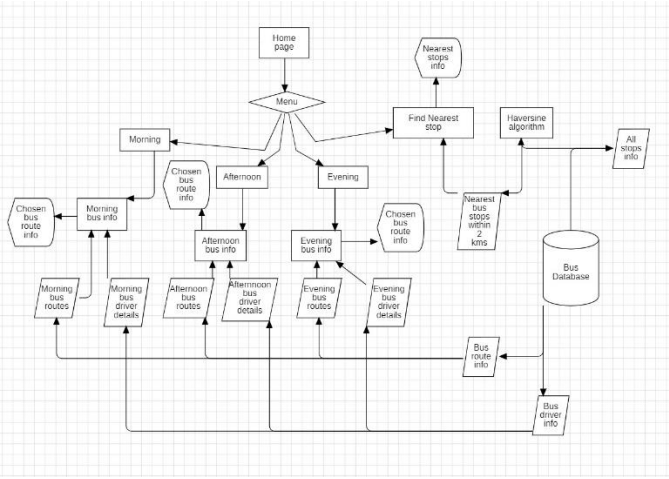


Fig 1. Process flow of the proposed system

The user interface is a web app designed using react where the user can select whether they want to find the stops which are closest to their location or view all the routes and their respective stops based on their time slots. The React web app is connected to the Node js and MySQL backend through Axios .

The database is hosted in a cloud environment so that based on the amount of system traffic, the processing power required to process the data in real-time can be scaled automatically.

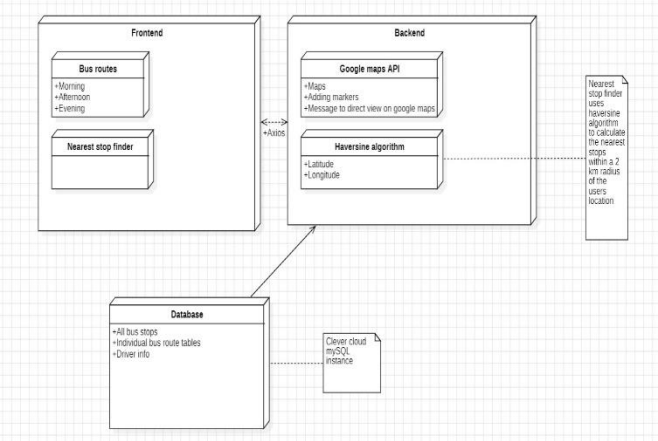


Fig 2. Architecture of the proposed system

As given in the Fig 2, the proposed model consists of frontend, backend and the database. A view of the database and its data type details are given in the Entity Relationship (ER) diagram as represented in the Fig 3. The bus routes are categorized based on the timing like morning, afternoon and evening.The user also has the availability of a feature to find the stop which are closest to them.In both the cases the web app uses axios library to fetch data from either the database incase of bus routes based on time slots or to fetch the data from the Haversine algorithm incase of finding the nearest stop information. The data which is returned in both the cases are the Route number,Stop name,Time and the Geospatial coordinates.

The workings of the components are described as below:

A. Google Map API

The Google Map API along with the '@react-google-maps/api' library is used to integrate Google Maps in the react app. Markers have been added to the map using the Marker component to show the exact location of the stops and on clicking the marker the user can view details of the Stop name and route number and a link which would redirect them to Google maps application with destination set as the stop using the InfoWindow component of the library. The geospatial coordinates of the stops to be marked in the map are sent from the database or Haversine algorithm to the Marker component so that it can be displayed on the map.

B. Haversine Algorithm

Haversine formula is used to implement an algorithm to calculate the individual distances between all the stops and the user defined location. The stops which are within an 2 KM radius of the user defined location are returned as the output of the algorithm.

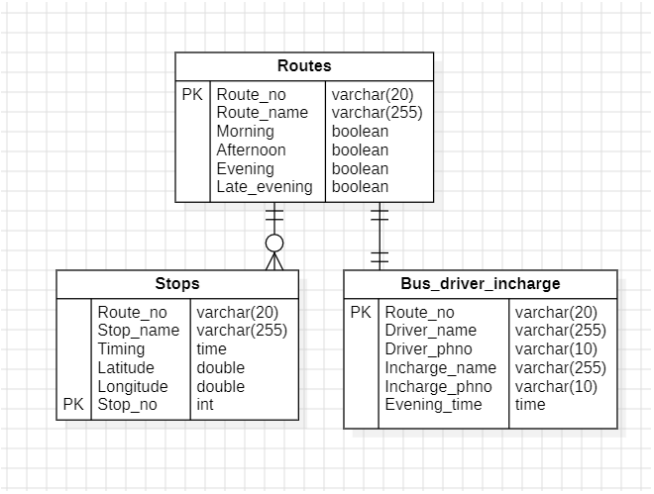


Fig 3. Entity Relationship diagram of the proposed system

The system uses three tables named Route (Route_no varchar, Route_name varchar,Morning boolean,Afternoon boolean,Evening boolean,Late_evening boolean) ,Stops (Route_no varchar,Stop_name varchar,Timing time,Latitude double,Longitude double,Stop_no int), Bus_driver_incharge where for each route there can be many stops(one-to-many relation) and for each route there will be a single bus driver (one-to-one relation).

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IV. Result

The model has been implemented and tested for various routes. A sample of the execution is depicted in the figure 4, where two locations have been suggested by the model. The figure 4 demonstrates that only Route No. 2 – Shanti Colony and Route No. 3 - Thirumangalam will be suggested based on the haversine formula.

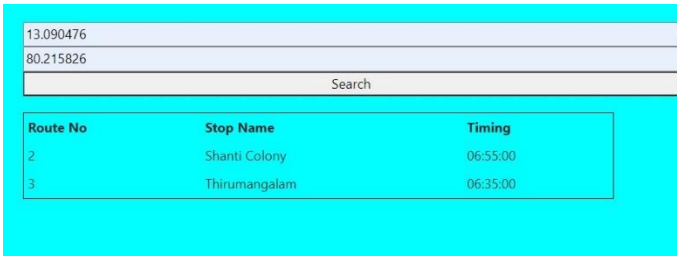


Fig 4. The route suggestion is done based on the haversine formula

The graph given in the figure 5, gives a graphical representation of the distance between the stops and the user defined destination. The distance measure is very less for the stops Shanthi Colony and Thirumangalam. Hence, these two stops are suggested.

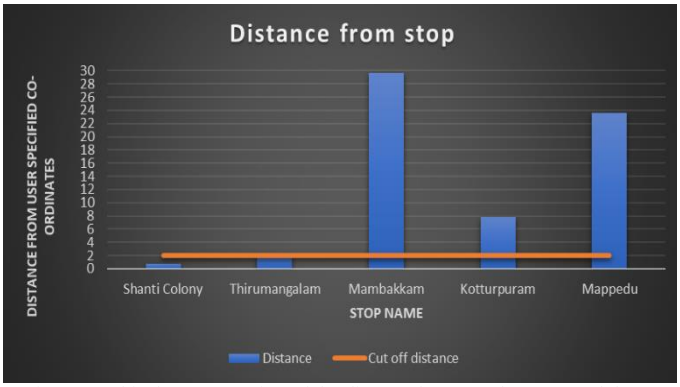


Fig 5: Graphical representation of the distance between the stops and the user defined destination

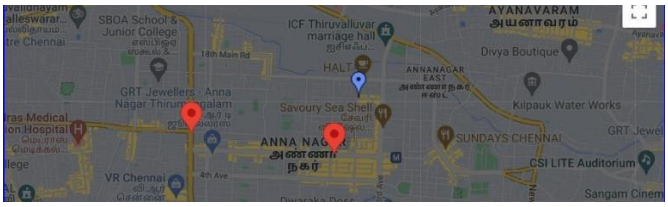


Fig 6: Representation of the Nearest Stops in Google Maps

Thus, the nearest possible locations are suggested to the user along with the route number. The arrival time at the destination is also predicted and given to the user. Based on these values, the user can select the bus route as per his convenience and board.

V. Conclusion

Thus, the proposed model serves as a comprehensive platform that provides the users with the detailed information. These details give the users information regarding various bus routes that are available in an educational institution. Along with it, the driver details for each route, the schedules are also given. The schedules are based on various time slots, that includes morning, afternoon shuttle, and evening buses. The main standout feature is the use of advanced geospatial functionality. This feature helps the users to effortlessly locate the nearest bus stop. The nearest bus stop is suggested based on the latitude and longitude coordinates provided.

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